

INFO-629 Applied AI Project Proposal

Project Title: Mental Health Text Classifier

Project Type: Practice Focused to build an application

Group Members:

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Project Description:

This project addresses the challenge of assessing and identifying potential mental health conditions reflected in a random unstructured text message. The goal is to develop a web-based application that accepts a block of text and applies a classification model to analyze the contextual pattern in the text and predicts which of the four categories it most closely represents: anxiety, depression, suicidal, or normal.

To accomplish this goal, the system will employ one of several classification models trained on labeled datasets from Kaggle, consisting of online posts and text messages annotated with these four mental-health categories. The expected outcome is a functional application capable of analyzing new text input and generating reliable, category-based, predictions that reflect the emotional or psychological state conveyed by the writer. This outcome can be used in mental health institutions to predict and resolve mental health issues, especially based on cyberbullying (solicited from Social media).

For this effort we will explore the use of several types of classification methodologies: a deep learning neural network and a multinomial naive bayes classifier. If the time presents itself we may explore other options as well.

Timeline:

What follows are the dates we expect to have each task completed by.

- February 7
 - Proposal Submission
 - Gather, Consolidate, and Clean Data
- February 12
 - Configure website host
- February 19
 - Create and Train Initial Classifier
- February 26
 - Backend Classifier Interaction

- Test and Iterate on Classifier
- March 5
 - Build website front end
 - Test and Iterate on Classifier
- March 12
 - Final testing of Classifier
 - Write Presentation Script
- March 19
 - Record Presentation
 - Final Submission

Resources:**Software and Libraries:**

- Python
- TensorFlow (Python Library) for building a deep learning neural network
- Scikit Learn (Python Library) for building probabilistic classifiers
- Streamlit (Python Library) to build a web interface

Possible Data Sources:

<https://www.kaggle.com/datasets/footsurebead/mental-status>

<https://www.kaggle.com/datasets/priyangshumukherjee/mental-health-text-classification-dataset>